

NovaPharma Iberia — Step 5 Review: Closure

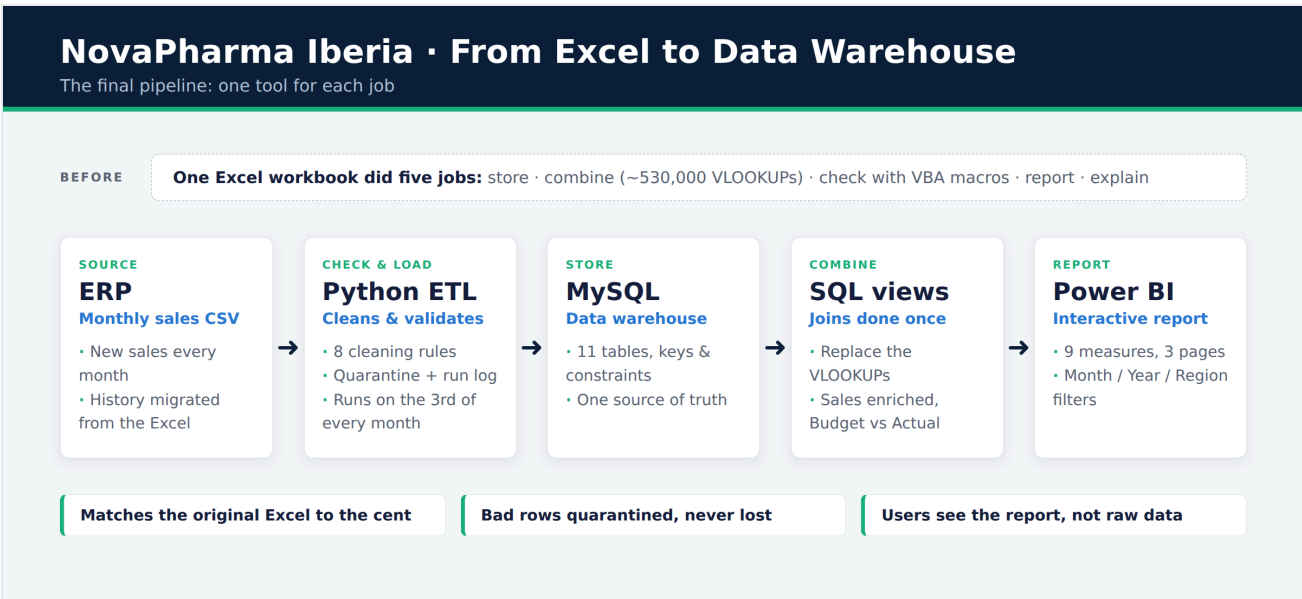
Digital transformation project · Excel → MySQL / Python / Power BI · Author: Nix · 29 September 2026

Starting point

In Step 0 we opened the original Excel file and found that it was doing five jobs at the same time: storing the data, combining it, checking it, reporting it and explaining it. Steps 1 to 4 took those jobs out of the workbook one by one. This closure puts the pieces side by side: what replaced what, and why the separation is better.

A. What replaced what

Job	In the Excel	Now
Store	Calendar, Products, Customers, Territories, Sales_Raw, Budget, Forecast, Market sheets	MySQL data warehouse: 11 tables (3 facts, 6 dimensions + quarantine and run log), with keys and constraints
Combine	Sales_Enriched: around 530,000 VLOOKUP formulas	SQL views: v_sales_enriched and v_bva. The joins are written once
Check & load	Data_Quality, Import_Staging, Macro_Log and VBA macros, run by hand	Python ETL: 8 cleaning rules, quarantine table, run log, Task Scheduler on the 3rd of every month
Report	Consolidation, BvA and KPI_Dashboard sheets	Power BI: 11 relationships, 9 DAX measures, 3 pages
Explain & configure	README sheet and Parameters (CurrentMonth...)	Repo README + About page. Parameters became slicers: the Month slicer replaces CurrentMonth



The final pipeline (image of post 5/5)

B. Why the separation is better

These were the biggest problems we found in the Excel, and what solves them now (the four in the LinkedIn post first):

Problem in the Excel	Now
Everyone had access to the raw data, so anyone could change a number by mistake and lead management to wrong decisions	Users only see the report. The data and the model stay protected in the warehouse
One broken VLOOKUP could corrupt the report without anyone noticing (70 SKUs written in lowercase never matched)	Bad rows are refused and parked in quarantine: visible, and never lost
Several versions of the truth: masters typed by hand, budget in another file	One source of truth, the warehouse. Excel total – quarantine = SQL total, to the cent (442.94M validated in Power BI)
A manual monthly refresh that only one person understood	The pipeline runs by itself, logs every run, and can be repeated without duplicating data

Honourable mentions, the ones that did not fit in the post:

- **Weight.** 17 sheets, 626,000 formulas and 6 VBA modules made the file slow and fragile. Now the database stores and combines, and Power BI only shows.
- **Forecast history.** Each quarter the forecast was overwritten. Now every version is kept in `fact_plan`, next to the budget, because they share the same grain.
- **Static vs interactive.** The dashboard was a fixed sheet sent as screenshots. Now the user filters by month, year and region and drills from brand to SKU.
- **Measures, not cells.** KPIs are 9 DAX measures that follow any filter, instead of formulas hidden in cells that only work for one layout.
- **Readable by anyone.** Code, model, report (PBIP as text files) and one review per step are documented and public on GitHub, not locked in one person's head.
- **Security.** The database password lives only in an environment variable, never in the code, the config file or Git.

C. The project, step by step

Step	What	Result
0	Excel as-is	Sheet dictionary. Five layers found mixed inside one workbook
1	Reverse engineering the model	Type, grain, keys and measures of every table
2	SQL data model in MySQL	Constellation schema: 3 facts, 6 dimensions. DDL + ER diagram
3.A	Python pipeline prototype	Profiling (45 duplicate IDs, 70 lowercase SKUs, 18 rows with placeholder code C9999) → 8 rules → quarantine → reconciliation to the cent
3.B	Production script	Rules as functions, idempotent monthly load of the ERP CSV, Task Scheduler on the 3rd
4.0–4.C	Power BI	Decide what to show → SQL views + 11 relationships → 9 measures → 3 pages with one theme
5	Closure	This review, the final pipeline, and the LinkedIn series closed

D. What stays open

Known limits, kept on purpose in the backlog and out of scope for this project:

- Re-running the same file re-inserts its quarantine rows (the facts are not duplicated, the quarantine log is).
- The ETL connects as root; a dedicated `etl_user` with minimum rights would be the next step. It could also pick up the newest CSV by itself.
- MySQL runs locally because no cloud platform was available; the design would move to Snowflake, BigQuery or Azure unchanged.
- Forecast versions only exist for 2026 (synthetic data).

E. The LinkedIn series

Post	Topic	Link
1/5	Step 0: the strategy	linkedin.com/feed/update/urn:li:activity:7508117104609390592
2/5	Steps 1 & 2: model and SQL	linkedin.com/feed/update/urn:li:activity:7508830718685483009
3/5	Step 3: Python ETL	lnkd.in/p/eg_6SF7V
4/5	Step 4: Power BI	lnkd.in/p/efKvYxBp
5/5	Step 5: closure	lnkd.in/p/eSHCNJ2Z

Result: the Excel is obsolete. Each job now lives in the tool built for it, and the data is handled correctly from its source to its visualization, which lets us draw more accurate conclusions from any business analysis.